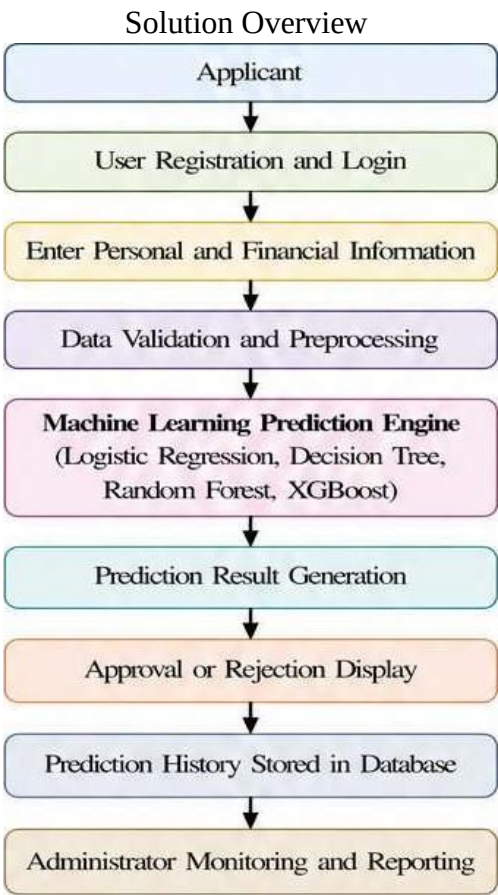


Date	03 July 2026
Team ID	SWTID-2026-9440
Team Size	4
Team Leader	Bodanapu Sai Krishna Reddy
Team Member 1	Anand Ch
Team Member 2	Singam Hemasree
Team Member 3	Damavarapu Sanjana
Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Proposed Solution

Introduction

The proposed solution is a Machine Learning—Based Credit Card Appmval Pædiction System that automates the evaluation Of credit card applications using historical financial data and pædictive analytics. The system minimizes manual intervention by accurately classifying applicants as eligible or ineligible for credit card approval. A web-based application built using Flask pmvides an intuitive interface for applicants and administrators, while Machine Learning models deliver fast, consistent, and reliable predictions.



Key Feattnæs of the Pmposed Solution

		Description
1.	Secure User Authentication	Allows applicants and administrators to register and log in securely using authenticated cœ&ntials.
2.	Applicant Data Collection	Collects personal, employnEnt, financial, and credit-related information through an intuitive web interface.
3.	Data Preprocessing	Handles missing values, enccx:les categorical variables, validates user inputs, and prepares data for Machine I._arning prediction.
4.	rvlzwhiœ Learning Prediction	Uses trained classification such as Regression, litcision Random Forest, and to predict cœdit
5.	Prediction Result	Displays the appmval or rejection result instantly along with the predicted eligibility status.
6.	History	Stores applicant details and prediction outcomes securely for future reference and administrative analysis.
7.	Administrative Dashboard	Allows administratol* to review prediction records, monitor application activity, and generate reports.

Advantages Of the Proposed Solution

Automates credit card approval prediction.

Reduces manual processing

- Improves prediction accuracy.
 - Supports secure applicant data management.
 - Ensures consistent decision-making.
 - Enhances customer experience.
 - Provides scalable web-based deployment.
 - Enables future Machine Learning improvements.

Conclusion

The proposed Machine Learning—Based Credit Approval Prediction System integrates modern web technologies with advanced Machine algorithms to automate the credit approval process. The solution improves decision accuracy, reduces operational workload, enhances customer satisfaction, strengthens data security, and provides a scalable platform suitable for deployment in real-world banking environments.